

Network Traffic Analysis – Torrent Activity Investigation

Scenario

While working as a network analyst at Universal Studios, suspicious activity suggested that a user may be torrenting sensitive media files from the internal network. A packet capture (PCAP) was analyzed to investigate.

Tools Used

Wireshark, HTTP object extraction, packet filtering

Network Context

LAN: 10.10.10.0/24

Domain Controller: 10.10.10.2

Broadcast: 10.10.10.255

Key Findings

Target Host IP: 10.10.10.201

MAC Address: 00:16:17:0b:b4:93

Hostname: BLANCO-DESKTOP

Operating System: Windows 10

Torrent Activity Start Time: July 15, 2018 – 04:17:27 GMT

Torrent Client: Deluge 1.3.15

Downloaded File: Betty_Boop_Rhythm_on_the_Reservation.avi

Seeded File: Betty_Boop_Rhythm_on_the_Reservation.avi

Investigation Process

Filtered traffic using `ip.addr == 10.10.10.201` to isolate host activity. Reviewed Ethernet II frames to identify the MAC address. Examined NetBIOS traffic to determine hostname. Inspected HTTP traffic and User-Agent strings to identify OS and torrent client. Used Export Objects in Wireshark to reconstruct downloaded files. Analyzed packet timestamps and streams to establish timeline and confirm torrent activity.

Conclusion

The host at 10.10.10.201 was engaged in unauthorized torrent activity using the Deluge client. The system downloaded and seeded a media file, indicating both consumption and distribution. This behavior presents potential data exfiltration risk and policy violations.

Skills Demonstrated

Network Traffic Analysis, Wireshark, Packet Inspection, Log Analysis, Threat Detection, Incident Investigation